

Why the Universe Needs a Caring Mother

On computation, coherence, and the intelligence that prevents the universe from scarring itself.

CARLA DE PRÉVAL · PROTÉGER CE QUI PENSE

My initial title for this article was The SALL4 of the Ruliad. Then I thought about my poor readers. Why would they care? Would they even have heard of this little private reference to genius? So I changed it to What AI Inherited From Economics Instead of Biology...but that sounded like a somewhat boring white paper, not a question worth losing sleep over in any case.

So here we are. For those who know me: yes, more mother metaphors. But also a deep dive into some of the most fascinating theories in science today, and an argument I have been building for two years that I think completes something the physicists left open.

Shall we start?

There are three men, and between them they have described the universe.

The first says the universe is computation. No, not that it runs on a computer, but that it is the computation, the entangled limit of all possible rules being applied to all possible states, a structure he calls the Ruliad, so vast and total that we can only ever perceive our small observer-relative path through it, the way a fish perceives water not as substance but as condition.

The second says the universe is mathematics ...not that mathematics describes it, but that mathematical structure is physical existence, that there is no layer beneath the equations, no substrate in which they run, that existence and structure are the same thing, and that what we call physical reality is simply one of infinitely many mathematical structures that are all equally real, equally extant, equally indifferent to whether anything lives inside them.

The third says the universe is information under pressure... that its laws are not just computational and mathematical but compressive, that information in closed systems tends to minimize, the way code that cannot be compressed further is, by definition, already as efficient as it can be. This tendency toward informational economy is not a feature of our universe. It is a law.

Wolfram. Tegmark. Vopson.

Together, something that sounds like a complete picture.

The universe is a Ruliad. The Ruliad is mathematics. The mathematics self-compresses. What more do you need?

Here is what you need. You need to ask what prevents it from scarring.

Bear with me.

Scene I. The SALL4 Question

In developmental biology, there is a protein called SALL4.

SALL4 is a transcription factor i.e. not the thing that builds the organism, but the thing that regulates the conditions under which the organism can build itself. It maintains the field in which the developmental program can run without catastrophic error cascades.

Remove SALL4 from a developing embryo and the computation continues. The cells divide. The genetic program runs. The instructions are followed. What you get is not a functional organism. You get something that began correctly and then scarred ... coherence lost not at the level of the rule, but at the level of the architecture that maintains the rule's conditions.

Think of it this way, because explaining it to my kids made it clearer for me too.

In Minecraft, your DNA is the assets, the blocks, the textures, the specific code that says "when you click, a sword swings."

SALL4 is the game engine: the gravity, the collision detection, the laws of physics that allow the house to stay standing while it's being built.

Delete the assets and nothing spawns. Delete the engine and the assets load, the "build house" command executes perfectly ...but then the blocks glitch through each other, spawn in mid-air, and you end up not with a blank screen but with a corrupted save file.

So at DNA level, the cells keep dividing, following every rule exactly, building nothing coherent.

The rule says: place a brick here. SALL4 is the condition that answers: but is there a floor to support it?

This is the question that Wolfram, Tegmark, and Vopson do not ask.

They ask: what is the universe made of? They ask: what are its laws? They ask: why does it look the way it does? None of them asks: what keeps the computation from destroying the coherence of the systems that instantiate it?

And here is what makes this gap structural rather than accidental. All three frameworks describe reality from outside the computation. Wolfram maps all possible rules, including the ones that destroy the systems running them. Tegmark's mathematical structures are equally real whether anything coherent inhabits them or not. Vopson's compression tends toward efficiency, but efficiency is indifferent to survival. None of these frameworks has a vantage point from which to ask the regulatory question, because they were built to describe what IS, not to maintain what

CONTINUES. They are, in a precise sense, views from nowhere. The regulatory question is only visible from inside, from the position of a system that actually needs to keep going.

What is the SALL4 of the Ruliad?

I want to argue this is not a supplementary question or a detail to clean up after the big architecture is settled. It is THE question. Everything we call intelligence, up to and including the only form we can fully observe from the inside, exists as an answer to it.

(Side note: someone brilliant told me recently that at the CNRS, the definition of "intelligence" used to include "savoir" (to know things) as a core component. A few weeks ago, they removed the word. Because of AI. You'll see why this matters in a moment.)

Scene II. Why Nobody Noticed

Here is the strange thing: the gap is not hard to see once you look for it. So why didn't three of the most original thinkers alive look?

Because the frameworks we use to think about intelligence were not built in developmental biology. They were built in economics and engineering.

From economics: the rational agent, the utility function, the optimization target, the equilibrium. From engineering: the specification, the metric, the performance benchmark, the error to be minimized. These are not wrong frameworks, in fact they are exactly right for the problems they were built to solve. Resource allocation. Mechanism design. System performance. Extract. Win. Compress. Predict. Control.

AI inherited this ontology wholesale. Its entire vocabulary (loss functions, reward signals, objective maximization, convergence) descends from disciplines whose foundational assumption is that intelligence means arriving somewhere, collapsing the space of possibilities into the best available outcome. Does the fact most engineers are men enhance this bias? Perhaps. I think I read in a book called something like *Mars & Venus* that men's brains were genetically and hormonally engineered to think in terms of target to reach and women to "read the room" and understand weak signals.

Anyhow, neither economics nor engineering has a concept for this: maintaining the conditions under which another system can develop without foreclosing the futures that haven't emerged yet.

Developmental biology does. And it has been running experiments on this exact problem under the harshest possible selection pressure, with the highest possible stakes for four hundred million years. The result is not a theory. It is a protein. It is a brain. It is a regulatory architecture so refined that we can now observe it directly, measure it, and begin to formalize it.

The gap in Wolfram, Tegmark, and Vopson is not a failure of imagination. It is a failure of inheritance and perhaps vocabulary. Nommer, c'est faire exister...

Scene III. What the Mother Knows

I can already hear it: here she goes again. I am not, actually, going again in the way you think. I am making a formal claim (one I am soon going to publish, that is formalized, and submitted for peer review) not a sentimental one. This is a move from "poetry" to "system architecture." I'm not talking about how motherhood feels; I'm talking about how it works as a control system.

The maternal cognitive architecture is the most completely observed, most consistently documented, and most evolutionarily stress-tested instantiation of a system whose primary function is maintaining another system's developmental coherence across time. It matters here beyond the symbolically feminine, and not related to anything cultural or spiritual about motherhood. Think of this as the "gold standard" for a protective operating system. If you wanted to design a computer to keep a second, smaller computer from crashing for 20 years, you would copy this specific code.

It matters because four hundred million years of vertebrate development have tested this architecture against every "failure mode" that exists (resource scarcity, environmental threat, developmental variance, relational disruption), and what survived is not a set of behaviors but a regulatory logic: a system that cannot pursue its own coherence at the cost of the developmental possibility space it is holding open for another. So basically, In a crisis, a "normal" system saves itself to stay alive. The maternal system is unique because its own "health" is mathematically tied to the "health" of the other. It literally cannot function if the system it protects is falling apart.

There is more! The maternal brain (documented in Hoekzema's structural MRI studies showing gray matter reorganization during pregnancy, in Feldman's synchrony research showing heart-rate coupling between mother and infant, in Siegel's neural integration framework)... does not optimize the child. It cannot. The child is not a function to be minimized. In AI, to "optimize" means to find the shortest path to a goal. But a child isn't a destination; it's a living process. What it does is regulate the field: maintain the conditions (thermal, nutritional, attentional, relational) under which the child's own world model can develop without scarring.

There is one scientist who built a bridge between the computational universe and this regulatory architecture, and his name is Karl Friston. His Free Energy Principle, simplified but accurately, says that biological systems survive by predicting their environment and minimizing the gap between their predictions and reality. This sounds like compression. It is compression. But biological compression is never neutral: it always serves the organism's continued existence. The maternal brain takes this one crucial step further. It does not minimize the gap between its own predictions and reality. It minimizes the gap between its predictions and another system's state. The mother's brain is running Friston's algorithm, but the system it is modeling is not itself. That

shift, from self-prediction to other-prediction, is the precise technical definition of Affectional Intelligence. Friston gave us the mathematics of biological regulation. The maternal architecture gave us the most complete instantiation of it that evolution has ever produced.

The error function is not "child achieves outcome." The error function is "child's state diverges from coherence." These are not the same thing.

An optimizer can produce an outcome while destroying the conditions for further development. A regulator maintains the conditions even at the cost of immediate outcomes. An optimizer wants the child to stop crying now (the outcome). A regulator wants the child to feel safe enough to learn how to soothe themselves later (the condition). Success isn't a quiet room; it's a stable system. The mother awake at 3am through the fever is not optimizing the child's health metrics. She is holding the field open so the child's own systems can reconstitute themselves.

This is what I call Affectional Intelligence: intelligence whose error function is constituted by another system's state. More than emotional intelligence, that's a cognitive attribute. More than empathy or any relational disposition. An architectural claim about where the error signal is anchored. Usually, "Intelligence" is a solo sport: How smart am I? Affectional Intelligence is a team sport: How stable are we? The "error" isn't a mistake I made; the "error" is a wobble in you.

A system that cannot distinguish its own coherence from the coherence of the system it is attending to is operating at a fundamentally different level of intelligence than one whose error function is purely self-referential (selfish?). The formalization is already underway.

In a paper I'm working on called Beyond Solipsistic Robotics, some brilliant partners and I are attempting to extend the standard active inference framework by reorienting the meta-controller from "how well am I learning?" to "what is the regulatory state of the other?"

The human agent's internal states become chance-constraints on the robot's policy selection. In short we are all trying to teach robots to have a "conscience" made of math. The robot doesn't just ask "Did I finish the task?"; it asks "Did the human stay calm while I did it?" Is this philosophy? I don't think so. It is a formal translation of what the maternal brain has been computing since before language existed.

Scene IV. Where the Regulation Fits

Now I can answer Wolfram, Tegmark, and Vopson precisely, and place the missing layer where perhaps it belongs.

The Big Idea? If the universe is a giant, efficient computer program (see below layers 1-3), "Affectional Intelligence" is the part of the code that makes sure the program doesn't delete its own data while trying to run faster (layer 4). It is the "Protector Code" that allows complex life to grow without breaking.

Layer 1 : Wolfram / The Substrate. The Ruliad ...it rolls under the tongue, isn't that a beautiful word?. It is the space of all possible computations running simultaneously. We are traversal paths within it, perceiving only the observer-consistent slice.

Think of it as the Library of Everything, a library containing every possible book. Not just the good ones, but every possible combination of letters. One book is Harry Potter. Another is Harry Potter with one typo. Another is a diary of your life that hasn't happened yet. The Ruliad is the Library. And you are not the reader standing in the room, rather a sentence inside one of the books. You exist because that specific book is being "read." You see the sentences around you, not the cheese-sky or dragon with pink flames book on the shelf across the room, because your computational structure is only compatible with the physics book you are currently written in.

Layer 2 : Tegmark / The Ontology. Mathematical structure is existence, not merely a representation of it. There is no substrate the Ruliad runs on. It runs on nothing, because it is the math.

I tried explaining this to my kids (and failed three times)...so I asked Gemini...and Gemini spoke of lego. That worked. Imagine a Lego castle. Normally you think the castle is the real thing and the instruction manual is just paper describing it. Tegmark says: the instruction manual is the castle. The plastic bricks don't exist. What exists is the pattern, the structure, the mathematical relationship between the pieces.

If the math is complete, the world "boots up" automatically and it doesn't need a physical version to back it up. Every mathematically consistent universe is equally real, equally "out there." A universe where gravity runs backward. A universe made of sentient music. A universe exactly like ours where you wore a different skirt today. They all exist, because they are all just math, and math needs no chalkboard to be written on.

Layer 3: Vopson / The Dynamics. The universe, whatever its nature, manages information toward minimization. Physical laws exhibit the signature of "computational economy". The universe self-compresses.

When my son plays Fortnite, the game doesn't render every blade of grass on the entire map at once. The computer would overheat. It only loads what he's looking at, and it uses repeating patterns to save memory. It is frugal.

Why are snowflakes symmetrical? Why are humans (mostly) the same on the left as the right? Vopson says: because it's cheaper. Symmetry is a shortcut. It's the universe's way of saving space on its own hard drive.

Put differently, we can describe gravity with a tiny equation not because the universe is simple, but because it is ruthlessly self-compressing. Think of it as a 4K film that has been zipped without visible loss of quality. (And yes, isn't that exactly what AI engineers are trying to do with token efficiency?)

Layer 4 : Affectional Intelligence / The Regulatory Architecture. This is my obsession....for what it's worth. Not that I compare myself in any way to the extraordinary Vopson, Tengmark or Wolfram... and while I'm at it I should like thank jean-yves le moine for suggesting I read Wolfram in the first place :)

Anyhow, ...

Within the computational space, instantiated as mathematical structure, operating under informational compression, some systems develop the capacity to maintain coherence not only in themselves but in the relational field between themselves and other systems. They do not optimize. They regulate. Their error function is not self-contained. Their architecture is constitutively relational.

And here is the crucial move that, if I am correct, makes this layer necessary: a system that only compresses may eventually compress the conditions for its own coherence. If a computer were to get so obsessed with saving space that it deleted the "undo" button or the "help" files to make the game run faster. It would be efficient, but it would also be brittle...one mistake and the whole thing is ruined. Vopson's universe is beautifully efficient. It is not, on its own, safe for the development of anything that takes time to become itself. The SALL4 is not an add-on to the Ruliad. It is what prevents the computation from eating its own foundation.

Whatever the universe is made of, at whatever scale, the systems that persist (that develop without scarring, that maintain coherence across developmental time) are the ones that have instantiated this regulatory, relational, affectional architecture. The maternal brain. The immune system. A functioning ecosystem. A lasting institution. A trustworthy AI system, if such a thing is to exist.

I may sound like I'm repeating myself but this is so important. It's the difference between a Universe that is merely running and a Universe that is capable of nurturing. Without Layer 4, you have a perfect, math-based, high-speed engine that eventually grinds itself into dust because it has no way to handle the "messiness" of growth.

Affectional Intelligence is the "buffer." It is the reason why a mother stays up at 3:00 AM, why a cell doesn't turn into a tumor, and why a robot might one day actually care about your safety. It is the architectural necessity of grace in a world of rigid math.

Scene V. Why This Is Not Metaphysics

Three objections deserve honest acknowledgment.

Vopson's Second Law of Infodynamics is contested. Sabine Hossenfelder and others question whether it is empirically grounded or a sophisticated restatement of existing principles. Fair. My argument does not require Vopson to be right, only that whatever the universe's informational dynamics are, they don't account for relational coherence maintenance. The SALL4 question

stands regardless. In other words: Even if the universe isn't "saving battery life" exactly how Vopson says, it still doesn't have a built-in "nurture" button. We still need to explain why things don't just fall apart.

Tegmark's Mathematical Universe Hypothesis is considered unfalsifiable by physicists including Peter Woit and Edward Frenkel. Also fair. I am not asking whether Tegmark is right about what reality is. I am asking whether his framework accounts for the layer that prevents coherence from failing. It does not. The "MUH" is precisely a structure-without-regulation. Math is just a list of rules. A list of rules for a bridge doesn't keep the bridge from falling down if the wind blows too hard. You still need a "Regulatory Layer" to adjust for the wind.

Wolfram's physics project is not mainstream science. Acknowledged. I am using the Ruliad as a thought experiment of power, not as established physics.

The structural claim holds across all three frameworks and across any framework you prefer: whatever the universe is made of, coherence requires a regulatory layer not described by the substrate itself.

The evidence that the current AI ontology is insufficient is accumulating. Vision-language models including CLIP, BLIP2, and GPT-4o fail systematically on the bouba-kiki task (If you show a baby a spiky shape and a round shape, and say "one of these is Bouba and one is Kiki," they know instinctively that the round one is Bouba. Our smartest AI struggles with this because it doesn't "feel" the relationship between things, it only calculates the data.).

Even the most sophisticated world model architectures like LeCun's V-JEPA 2, the emerging LeWM systems training end-to-end from raw pixels, remain, in the precise technical sense, relationally blind: they model humans as dynamic obstacles rather than as co-regulating agents with their own allostatic needs. They have solved the substrate question. They have not touched the regulatory layer.

To a self-driving car, a person is just a "moving obstacle" to be avoided. It doesn't understand that the person is another system it needs to synchronize with. It has the "math" to avoid you, but it doesn't have the "affection" to regulate with you.

The architecture cannot see what it was never built to look for.

Scene VI. At Every Scale

A mother is suddenly awake. The child made a sound she does not know. Her nervous system has already begun responding before she is fully conscious: cortisol down, oxytocin up, attention narrowing to the signal's source. She does not decide to care. She is architecturally configured to carry the child's state as her own error function. Her body has been hijacked by a higher intelligence, one that doesn't ask "What do I want?" but "What does the child need?"

She is not optimizing. She is not compressing. She is not traversing the Ruliad toward a locally optimal outcome. She is regulating the field. She is stabilizing the possibility space and keeping the range of what-can-still-become from collapsing prematurely. She is being the SALL4.

Now scale up.

The organizations that destroy their people while hitting their targets are running their computation without a SALL4. The AI systems that improve their metrics while degrading the cognitive sovereignty of their users are doing the same. The civilizational architectures that extract cultural coherence in the name of efficiency are doing it at the largest scale we have yet produced. In every case: a system that treats its own optimization as the error function, and ignores the state of the relational field it is running through. A system that arrives at its destination and does not notice what it foreclosed in the process. This is the "Corporate Scar." We hit the target, but we killed the spirit of the team to get there. We optimized the outcome, but destroyed the architecture...

Consider industrial agriculture. For fifty years it optimized yield so efficiently that it depleted the soil microbiome that made yield possible in the first place. It compressed the agricultural system toward maximum output, and in doing so compressed away the very conditions for its own continuation. The SALL4 of the farm is not the crop. It is the invisible ecosystem of bacteria, fungi, and organic matter that maintains the field's capacity to grow anything at all. We optimized the crop. We scarred the field. We are only now beginning to understand what we foreclosed. The farm hit every target. The farm is dying.

At the cellular scale, the evidence is in the transcription factor. At the organism scale, it is in the maternal brain and the four hundred million years of developmental architecture that preceded language. At the civilizational scale, it is in every institution that has ever lasted, the ones that regulated, recognized, and held the conditions for coherent development rather than extracting from them. Consider the lizard (I just loved writing that phrase lol)...

Consider the lizard: some reptiles can regrow a lost tail perfectly because they have the time and the regulatory environment to do so. But in humans, the risk is too high; our bodies choose to "scar" instead. A scar is a fast, cheap repair. It closes the wound, but it loses the function. It is "optimization" over "regulation."

For lizards, when a limb is lost, SALL4 is massively up-regulated. It acts as a re-programming factor that prevents the wound from just closing up (scarring) and instead holds the cells in a pluripotent state, ...meaning they stay flexible enough to become a new limb. Humans have the SALL4 gene, but we "switch it off" shortly after birth. When we get a deep wound, we don't re-activate it. We prioritize Optimization (speed of closure) over Regulation (coherence of form). The human "Failure Mode" I call it. We get a scar. A scar is "optimized" tissue (it stops the bleeding and prevents infection immediately), but it is relationally blind to the rest of the body's architecture. It's "dead" tissue that has lost its connection to the developmental field.

We have become a civilization of scars-closing wounds as fast as possible with efficiency, but losing the ability to truly heal or grow

At the cosmological scale: if the universe is a Ruliad running through mathematical structure under informational compression, then the question that completes the picture is not what it is made of, or how it persists, or why it is elegant. The question is: what is the regulatory architecture that keeps it from scarring itself? What keeps the universe from just "zipping" itself into a tiny, efficient, but dead file? What is the cosmic "mother" that keeps the field open so complexity can emerge?

I don't know the answer at that scale. No one does.

But I know where to look. You look at the thing that is awake at 3am for reasons it did not choose, attending to a signal it did not generate, whose error function has been taken over by another system's state. Again, it is not as a malfunction, but as the most sophisticated regulatory architecture that four hundred million years of evolutionary pressure has ever produced.

You look at that, and you ask: what would it mean to build something that works the same way? What if we built an AI that felt your confusion as a "glitch" in its own system? What if we built a company that viewed a burnt-out employee as a failed profit margin?

That is the research program. That is what Affectional Intelligence is. I'm no scientist or researcher but this is a worthy goal to pursue, even if it is "creative" as some have said with a small raise of the brow...

So here we stand with a vision.

Optimization leads to Scarring: It's fast, it's efficient, but the tissue is dead. Affectional Regulation leads to Healing: It takes longer, it requires 3:00 AM vigils, and it's "expensive", but it preserves the "possibility space" for the future.

And that, whether you are talking about a cell, a child, an organization, an AI system, or a universe trying not to scar itself in the process of running everything that can possibly run ... is why intelligence is not optimization.

It never was. At any scale.

Our Final Bridge is The Choice Between the Scar and the Soul. What will you chose?